Design of a Comprehensive Web-Based Academic Management Platform with Integrated Intelligent Features as an Alternative for Traditional LMS

Diploma Thesis

By

Olzhas Karabekov

Askat Narinbetov

Nurislam Baltabekov

School of Software Engineering
Astana IT University

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Supervisor: Zhaksylyk Kozhakhmet

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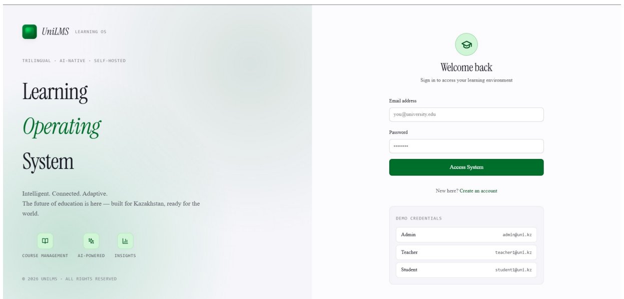
Technical Overview

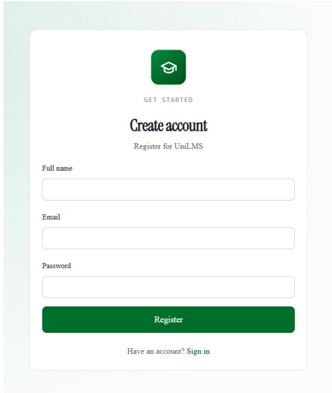
The diploma project resulted in the development of **UniLMS** — a comprehensive web-based academic platform built as a modern alternative to traditional learning management systems such as Moodle, Canvas, and Blackboard. The platform was designed around three principles: a contemporary user experience, native artificial-intelligence integration in core academic workflows, and complete trilingual coverage in English, Russian, and Kazakh. The system is delivered as a modular monolith with three clearly separated layers — presentation, application, and data — and runs as a three-service Docker Compose stack.

The presentation layer is built with **Next.js 14** using the App Router, **Tailwind CSS** for utility-first styling, **TanStack Query 5** for client-side server-state management, and **Framer Motion** for transitions and micro-interactions. A custom design system based on the OKLCH color space provides perceptually uniform gradients, dual themes (forest green for light mode and violet for dark mode), and a density toggle. The application layer is implemented in **NestJS 10**, which provides modular architecture, dependency injection, validation, role-based access control, and Server-Sent Events for sub-second real-time notifications. **PostgreSQL 15** together with **Prisma 5.22** forms the data layer, with fourteen domain models, ACID transactions, full-text search, and automatic migrations on startup.

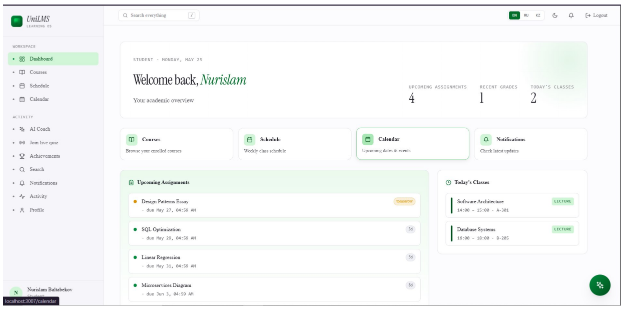
The intelligent layer of the platform is powered by **Anthropic Claude** and exposes five production endpoints: structured assignment feedback, topic-based quiz generation, course summarization with workload estimation, individual student performance analysis with risk scoring, and a streaming AI chat assistant. Every endpoint accepts a language parameter so the model responds in the user's preferred locale, and a built-in demo mode allows the system to run without any external API key. End-to-end testing is performed with Jest and Supertest, the project compiles in strict TypeScript mode with zero errors, and the entire stack ships with a single `docker compose up` command for reproducible deployment.



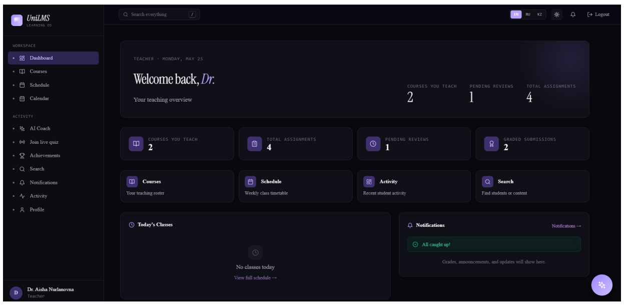
(a) Authentication screen with email/password login



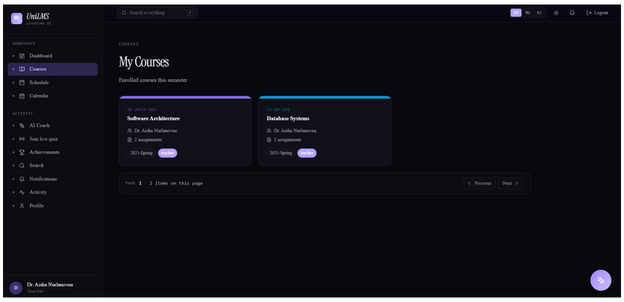
(b) Registration screen with role selection



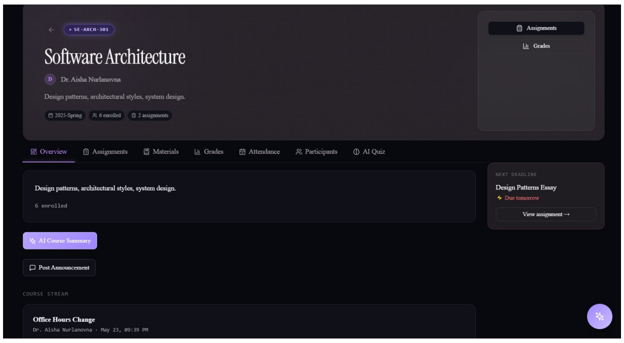
(c) Student dashboard with courses and deadlines



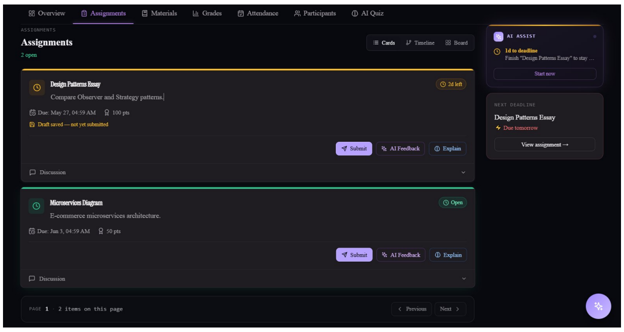
(d) Teacher dashboard with course overview



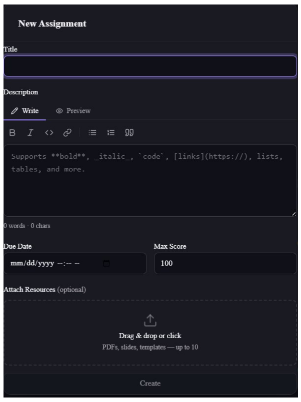
(e) Course catalog with search and filtering



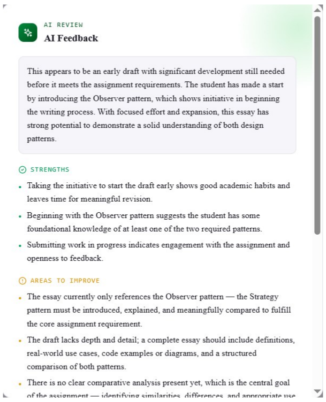
(f) Course detail page with materials and modules



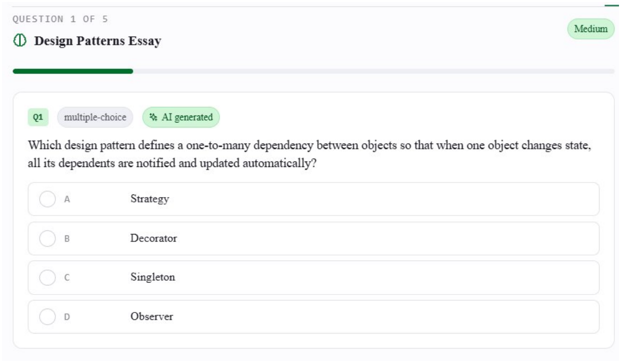
(a) Assignment view from the student perspective



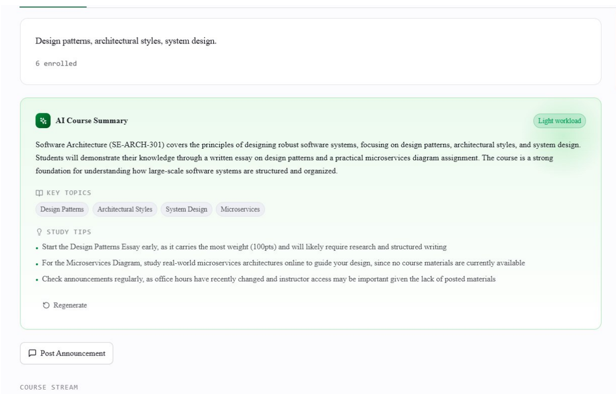
(b) Assignment creation by a teacher



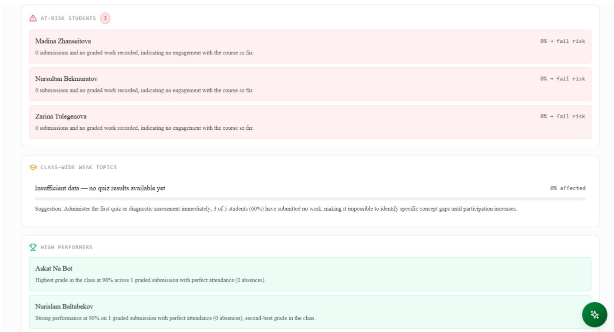
(c) AI-generated structured feedback on a submission



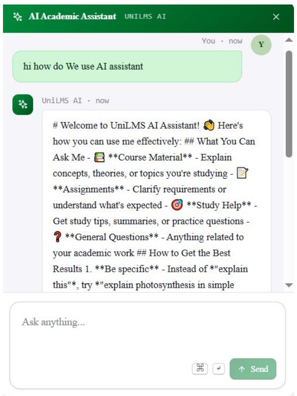
(d) AI quiz generation with language and difficulty



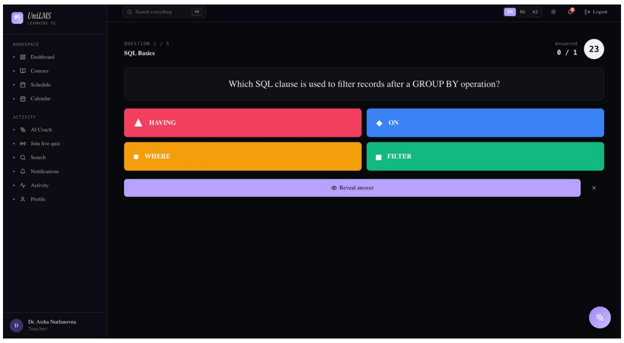
(e) AI course summary with workload estimation



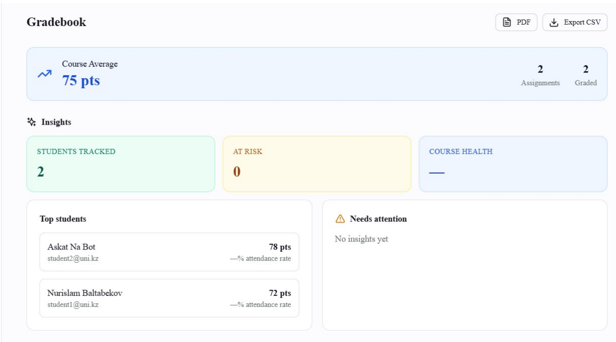
(f) AI Student Analysis with risk level and tips



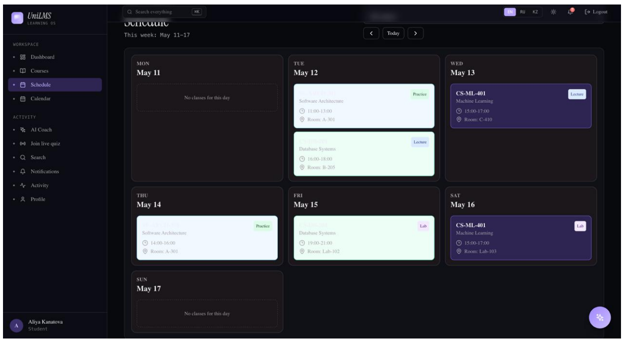
(a) Streaming AI chat assistant (SSE)



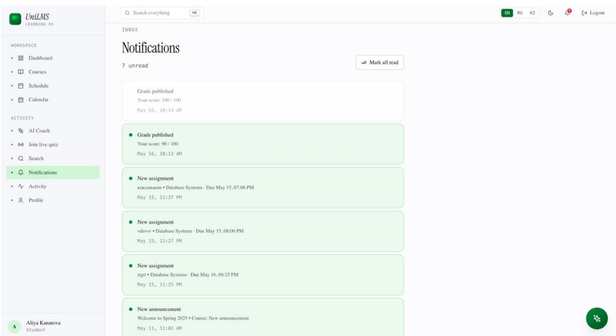
(b) Quiz module with Kahoot-style live mode



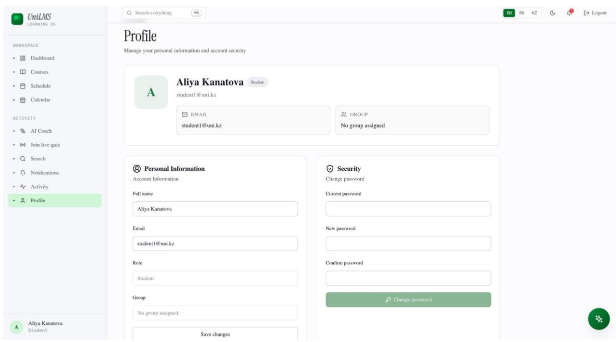
(c) Gradebook with assignment breakdown



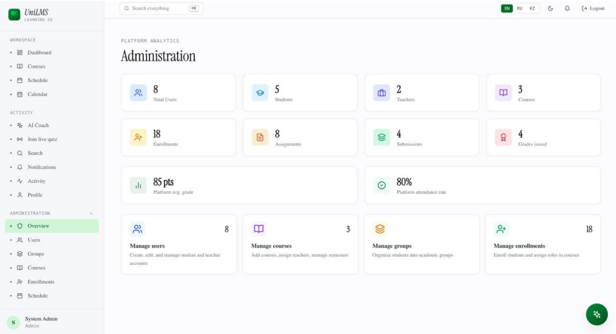
(d) Weekly schedule and calendar



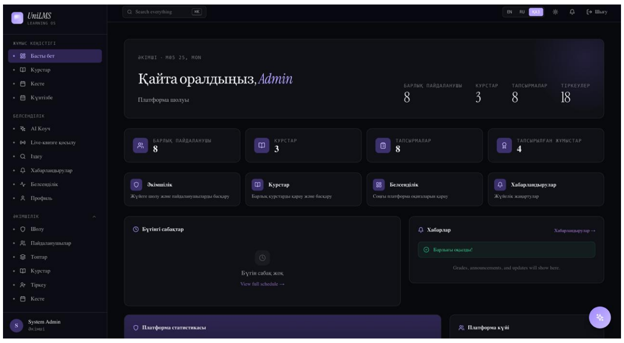
(e) Real-time notifications powered by SSE



(f) Profile and settings with language and theme



(a) Administrative dashboard with platform statistics



(b) Kazakh-language interface and dark (violet) theme